Textual Analysis in Finance

Week 2: Classic NLP in Finance - Sentiment Analysis and Topic Modeling

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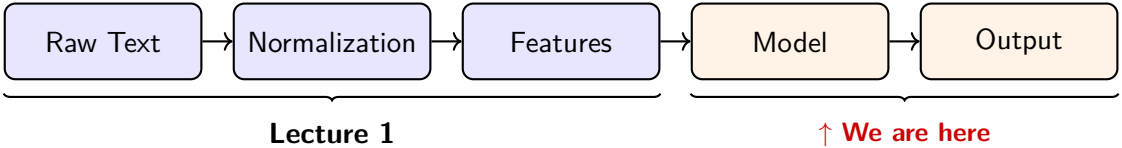
Where are we?


Raw Text
Normalization
Features
Model
Output
Lecture 1
↑ We are here

Sentiment Analysis

What is "Sentiment"?

- *"A thought, opinion, or idea based on a feeling about a situation, or a way of thinking about something"* (Cambridge Dictionary)

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- **Sentiment analysis** aims to identify the emotional tone expressed in a text (sentence, paragraph, or document), such as *enjoyment*, *anger*, *disgust*, *sadness*, *fear*, or *surprise*
- The most common type of sentiment analysis is to classify the *polarity* of a text, i.e., whether it conveys a **positive**, **negative**, or **neutral** tone

Sentiment analysis in finance

- Text polarity is commonly interpreted as the degree of *optimism* or *pessimism* (or positivity/negativity) regarding economic or financial outcomes
  - Firm managers' outlook on future firm performance
  - Central bankers' perspective on the current economy
  - **Our main focus in sentiment analysis**

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- From now on, *sentiment* refers to the positivity, negativity, or neutrality expressed in a text; and *sentiment analysis* refers to assigning a numerical score that captures this sentiment

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  - **Our main focus in sentiment analysis**
- From now on, *sentiment* refers to the positivity, negativity, or neutrality expressed in a text; and *sentiment analysis* refers to assigning a numerical score that captures this sentiment
- Textual sentiment has been shown to predict asset returns [Jiang et al., 2019, Loughran and McDonald, 2011, Tetlock, 2007], trading volume [Siganos et al., 2017], and financial volatility [Ahrens et al., 2025, Audrino et al., 2020]

Sentiment analysis methodology

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- Sentiment analysis can be conducted using both **supervised** and **unsupervised** (or dictionary-based) approaches
- **Supervised** approaches require labeled text data, which are often scarce or costly to obtain in economics and finance
- **Dictionary-based** approaches do not require labeled documents, but rely on pre-defined sentiment dictionaries to classify text polarity

Dictionary-based approaches - How to?

Two sentences: **the market is bullish** and **investors feel bearish about the market**

Step 1: Text normalization and vocabulary construction

s1: market bullish; s2: investor feel bearish market

$$V = [\text{“bearish”}, \text{“bullish”}, \text{“feel”}, \text{“investor”}, \text{“market”}]$$

Step 2: Construct the document-term matrix

$$\begin{array}{c} \\ s1 \\ s2 \end{array} \begin{array}{c} \begin{array}{ccccc} \text{bearish} & \text{bullish} & \text{feel} & \text{investor} & \text{market} \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 \end{bmatrix} \end{array}$$

Dictionary-based approaches - How to?

Step 3: Select a pre-defined dictionary that assigns polarity labels to words

$$\begin{array}{c} \\ s1 \\ s2 \end{array} \begin{array}{c} \begin{array}{ccccc} \text{bearish} & \text{bullish} & \text{feel} & \text{investor} & \text{market} \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 \end{bmatrix} \end{array}$$

Step 4: Compute the sentiment score

$$sentiment_d = \frac{\sum_{w \in d} [\mathbb{I}(w \in V^{Pos}) - \mathbb{I}(w \in V^{Neg})]}{l_d}$$

where V^{Pos} and V^{Neg} denote the pre-defined sets of positive and negative words, respectively; l_d is the total number of words in document d

Dictionary-based approaches - Remarks

- Several variants of the sentiment scoring formula exist; replacing l_d with $\mathbb{I}(w \in V^{Pos}) + \mathbb{I}(w \in V^{Neg})$, meaning focus only on sentimental words in d [Henry, 2008]

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- **Key takeaway:** The choice of the sentiment lexicon is the most important modeling decision in dictionary-based sentiment analysis

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- Instead of the document-term matrix, one may use the *tf.idf* representation to account for word importance
- The sentiment measure depends critically on the choice of V^{Pos} and V^{Neg}
- **Key takeaway:** The choice of the sentiment lexicon is the most important modeling decision in dictionary-based sentiment analysis
- Popular choices of the sentiment dictionary in finance: Harvard-IV [Stone et al., 1966], Henry's dictionary [Henry, 2008], and Loughran-McDonald dictionary [Loughran and McDonald, 2011]

Dictionary-based approaches - The choice of the sentiment dictionary

- Harvard-IV:
  - Designed at Harvard in the 1960s-1970s for psychological and social science research
  - Categories include: positive (1,915 words), negative (2,291 words)
  - Commonly used in the pioneering studies of textual analysis in finance [Tetlock, 2007, Tetlock et al., 2008]
  - Words like "liability", "capital", or "tax" are flagged as negative

[1] Notice: earnings press releases $\neq$ earnings conference calls

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- Henry's dictionary [Henry, 2008]:
  - The first sentiment dictionary for financial texts (earnings press releases)
  - A conservative dictionary with 85 negative words and 105 positive words
  - Words like "loss", "losses", "impairment" are missing; while "decrease" and its variants are flagged as negative
  - Price et al. [2012] claim Henry's dictionary is better than Harvard-IV at measuring the tone of earnings conference calls [1]

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Dictionary-based approaches - The choice of the sentiment dictionary

- Loughran-McDonald (LM) dictionary:
  - Developed by Loughran and McDonald [2011] to address the limitations of Harvard-IV (no financial context) and Henry's dictionary (limited coverage)
  - Constructed using U.S. financial disclosures (10-K and 10-Q filings)
  - 354 positive words and 2,329 negative words
  - Besides "Positive" and "Negative", there are "Uncertainty" ($\sim$ 300 words), "Litigious" ($\sim$ 900 words), "Constraining" ($\sim$ 180 words), and "Modal words" ($\sim$ 70 words)
  - The most widely used sentiment dictionary in empirical financial research
  - Using this dictionary, Loughran and McDonald [2011] show that stock markets react primarily to negative tone in firm disclosures, but not positive tone

Dictionary-based approaches - Pros & Cons

- Pros:
  - Easy to understand and implement
  - No labeled data required
  - Highly reproducible and stable, i.e., results are deterministic given a dictionary
  - **High transparency and interpretability**
    - Sentiment score is linked to word counts - You can always say: "This document is positive because "good" appears 100,000 times"
    - Easy to explain to regulators, referees, and practitioners

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- Cons:
  - Strong dependence on the chosen dictionary
  - Limited adaptability - Can we use the LM dictionary on central bank speeches?
  - At the end, it's a BoW method, meaning word order, word semantics, and contexts are completely ignored
  - **Sentiment is more complicated than word counts** → We talk about this now

Supervised approaches

Sentiment analysis can be viewed as a prediction problem:

$$y = X\beta + \varepsilon$$

where y is the sentiment of all documents; X is the document-term or *tf.idf* matrix

The dictionary-based approaches implicitly assume $\beta_w = 1$ for positive word, -1 for negative words, and 0 otherwise

Is it really that simple?

Supervised approaches - How to?

Core task: Given a text document X, assign it a sentiment score or label y.

We want to find a function, $f(\cdot)$,

$$y = f(X) + \varepsilon$$

that maps text, X, to sentiment, y

Steps (in order):

1. Make an assumption about $f(\cdot)$, e.g., linear or non-linear, parametric or non-parametric, a random forest or a neural network, etc.
2. Provide many documents with known sentiment labels (X_d, y_d)
3. Use these examples to estimate the mapping $f(\cdot)$
4. Once $f(\cdot)$ is learned, it can predict sentiment for new, unseen texts

Supervised approaches - Challenge

Main challenge: Documents with known sentiment labels are rare!

Common solutions:

- *Human annotation:* Researchers manually label text using domain experts or trained annotators
  - Examples include Li [2010], Malo et al. [2014], Huang et al. [2023], Pfeifer and Marohl [2023], Shapiro et al. [2022]
    (+) high-quality labels
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    (+) high-quality labels
    (–) time-consuming and costly
- *Using economic outcomes as proxies for sentiment:*
  - Jegadeesh and Wu [2013] and Frankel et al. [2022] use stock excess returns as labels (y), with 10-K filings as inputs (X)
  - Ahrens et al. [2025] use changes in forecasts of macroeconomic outcomes (GDP, CPI, unemployment) as labels (y), with the Fed's Greenbook as inputs (X)
    (+) inexpensive and economically meaningful labels
    (–) labels are noisy and conflate sentiment with other shocks

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  - Leverage the power of machine learning
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  - Leverage the power of machine learning
  - Usually, higher predictive power than dictionary-based approaches
- Cons:
  - Requires labeled data
  - Domain dependence - Models trained on news, predict on financial statements
  - Machine learning pitfalls - overfitting, data leakage, model choice, etc.
  - The extent of interpretability is lower than dictionary-based approaches
  - At the end, it is **still BoW**, meaning word order, semantics, contexts are **completely ignored**

Topic modeling

What is a "Topic"?

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- Examples of topics in financial text:
  - Monetary policy: *interest rate, inflation, tightening, target*
  - Firm performance: *earnings, revenue, growth, margin, guidance*
  - Risk and uncertainty: *risk, volatility, uncertainty, exposure, shock*

Topic modeling - Technical perspectives

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- Unlike standard dimensionality reduction, topic modeling aims for **interpretable and meaningful dimensions**, i.e., each dimension is a topic
- From another perspective, topic modeling can be seen as a form of clustering:
  - words with similar meanings tend to co-occur $\rightarrow$ form topics
  - documents are mixtures of these word clusters (topics)

Topic models - Latent Semantic Analysis

- If topic modeling is viewed as a **dimensionality reduction** problem, then applying **Singular Value Decomposition (SVD)** to the document-term matrix is a natural starting point

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Topic models - Latent Semantic Analysis

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- This idea underlies **Latent Semantic Analysis (LSA)** [Deerwester et al., 1990], also known as **Latent Semantic Indexing**
- **Steps:**
  1. Construct the document-term matrix (often using *tf.idf*), denoted by X
  2. Apply SVD:
  $$X = U\Sigma V^\top$$
  3. Keep only the top K **singular values** and the corresponding singular vectors:
  $$U_K,\ \Sigma_K,\ V_K, \qquad (X \approx U_K \Sigma_K V_K^\top)$$
  4. $V_K \in \mathbb{R}^{|V| \times K}$: **term-topic matrix** (importance of each term in each topic)
  5. $U_K \in \mathbb{R}^{N \times K}$: **document-topic matrix** (topic composition of each document)
  6. $\Sigma_K \in \mathbb{R}^{K \times K}$: **topic strength matrix** (importance of each topic in the corpus)

Latent Semantic Analysis - Pros & Cons

- **Pros - It's a linear algebra approach**
  - Conceptually simple and mathematically well-motivated
  - SVD is computationally efficient and scalable
  - No hyperparameters to tune, no training in the probabilistic sense

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  - No hyperparameters to tune, no training in the probabilistic sense
- **Cons**
  - **Not a generative model**
    - No explicit probabilistic story for how documents are produced
    - The SVD must be recomputed with new documents
  - Topics may be difficult to interpret
    - Term-topic (V_K) and document–topic (U_K) matrices can contain **negative** values
    - Topics are not guaranteed to correspond to coherent semantic themes
  - Implicitly relies on **Gaussian assumptions**
    - Word usage is treated as continuous and symmetric
    - Inconsistent with sparse, discrete word-count data

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- LDA is a **generative probabilistic model**: it explicitly models how documents are produced
- **Model intuition** - imagine you are an author writing an article:
  1. First, choose how much each topic will contribute to the article
  2. Then, begin writing; for each word position:
     - A topic is drawn from the document's topic mixture
     - A word is drawn from the chosen topic
  3. Repeat until the article is complete

Latent Dirichlet Allocation - The generative process

- **Notations**
  - We observe N documents, each document d contains N_d words
  - The vocabulary size of $|V|$
  - Assume a fixed number of K topics
- **Model flow**
  - Choose $\theta_d \sim \text{Dir}(\alpha)$, $d = 1, ..., N$; $\text{Dir}(\alpha)$ is Dirichlet distribution with parameter α
    - *Intuition:* choose the distribution of topics in document d
    - θ_d and α are K-dimensional vectors; $\alpha > 0$ and usually < 1 to capture the sparsity, i.e., a document doesn't contain all topics

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  - Choose $\varphi_k \sim \text{Dir}(\beta), \quad k = 1, ..., K$
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    - φ_k and β are $|V|$-dimensional vectors; $\beta > 0$ and usually < 1 to capture the sparsity, i.e., a topic doesn't contain all words
  - For each word j in document d
    - Choose a topic $z_{d,j} \sim \text{Multinomial}(\theta_d)$ $(z_{d,j} \in \{1, ..., K\})$
    - Choose a word $w_{d,j} \sim \text{Multinomial}(\varphi_{z_{d,j}})$

Latent Dirichlet Allocation - Why the Dirichlet Distribution?

- In LDA, we repeatedly (i) choose a **topic** for a word, and then (ii) choose a **word** given the topic
- This requires probability distributions over:
  - topics (for each document)
  - words (for each topic)
- Example: suppose document d discusses three topics: 60% Topic A, 30% Topic B, 10% Topic C, 0% Topic D, 0% Topic E
- The corresponding topic-proportion vector: $\boldsymbol{\theta}_d = (0.6,\ 0.3,\ 0.1,\ 0.0,\ 0.0)$
- $\boldsymbol{\theta}_d$ is **non-negative** and **sum to one** $\rightarrow$ It lies on a **simplex**
- The **Dirichlet distribution** is a natural distribution defined on the simplex

Latent Dirichlet Allocation - Estimation and Inference

- Parameters to estimate are α (document-topic) and β (topic-word)
- We choose α and β to maximize the likelihood of the observed words
- It is equivalent to maximizing the following likelihood function

$$p(\boldsymbol{w}_d|\alpha,\beta) = \int p(\theta_d|\alpha)\left[\prod_{j=1}^{N_d}\sum_{z_{d,j}} p(z_{d,j}|\theta_d)p(w_{d,j}|z_{d,j},\beta)\right]d\theta_d$$

- This likelihood is **intractable**, i.e., a closed-form solution for α and β doesn't exist $\rightarrow$ Blei et al. [2003] suggest using Expectation-Maximization estimation
- After estimation, we are also interested in the distribution of topics of a document d, i.e., $p(\theta_d, \boldsymbol{z}_d|\boldsymbol{w}_d,\alpha,\beta)$. This is again **intractable** and can be done using **variational inference**

Latent Dirichlet Allocation - Pros & Cons

- **Pros:**
  - More interpretable topics (than LSA)
  - Generative probabilistic model - Coherent story of how text is produced
  - Handles well new documents - Can infer topic proportions for unseen documents without retraining the entire model

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  - Handles well new documents - Can infer topic proportions for unseen documents without retraining the entire model
- **Cons:**
  - **Bag-of-Words underlying**
  - Computationally intensive (estimation, inference)
  - Uncovered topics can still be unstable and hard to interpret

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- By capturing the **semantics of words**, we can cluster semantically similar words to form topics (covered in the next lecture)

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- A **supervised approach** to topic modeling is also possible:
  - Analogous to sentiment analysis
  - The target, y, now represents topics instead of sentiment
  - The task becomes a **classification problem**

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We will do this in the assignment

Coming next

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- Hands-on practice
  - Sentiment analysis with Financial Phrasebank dataset [Malo et al., 2014]
  - LDA with central bankers' speeches
  - Notebook
- Next lecture
  - Neural embedding: equipping words with semantic meaning

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